

**SHET L.U.J. AND SIR M.V. COLLEGE**  
**SUBJECT :- CYBER SECURITY**  
**PRACTICAL NO: 8**

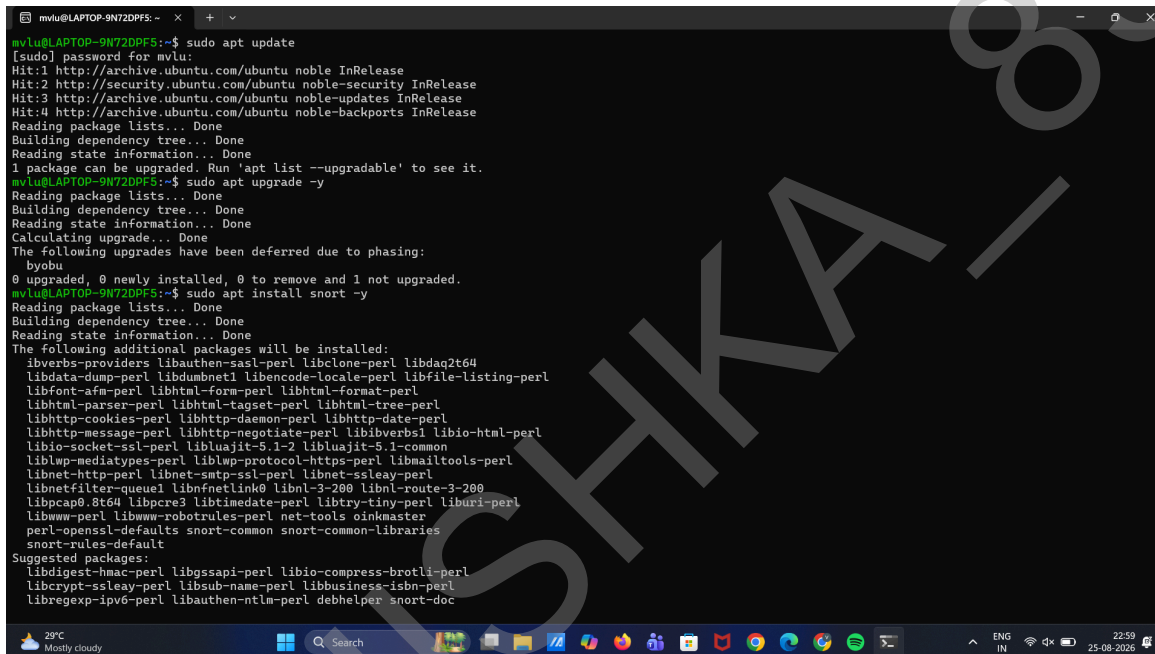
**AIM:-** Set up and configure an intrusion detection system (IDS) to monitor network traffic and detect potential security breaches or malicious activities.

**STEP 1.** Open the Terminal and run:

`sudo apt update`

`sudo apt upgrade -y`

`sudo apt install snort -y`



```
mvl@LAPTOP-9N72DPF5:~$ sudo apt update
[sudo] password for mvl:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
1 package can be upgraded. Run 'apt list --upgradable' to see it.
mvl@LAPTOP-9N72DPF5:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  byobu
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
mvl@LAPTOP-9N72DPF5:~$ sudo apt install snort -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ibverbs-providers libauthn-sasl-perl libclone-perl libdaq2t64
  libdata-dump-perl libdumbnet1 libencode-locale-perl libfile-listing-perl
  libfont-afm-perl libhtml-form-perl libhtml-format-perl
  libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl
  libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl
  libhttp-message-perl libhttp-negotiate-perl libibverbs1 libio-html-perl
  libio-socket-ssl-perl liblua5.1-2 liblua5.1-common
  liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl
  libnet-http-perl libnet-smtp-ssl-perl libnet-ssleay-perl
  libnetfilter-queue1 libnfnetlink0 libnl-3-200 libnl-route-3-200
  libpcap0.8t64 libpcrc3 libtimedate-perl libtiny-perl liburi-perl
  libwww-perl libwww-robotrules-perl net-tools oinkmaster
  perl-openssl-defaults snort-common snort-common-libraries
  snort-rules-default
Suggested packages:
  libdigest-hmac-perl libgssapi-perl libio-compress-brotli-perl
  libcrypt-ssleay-perl libsub-name-perl libbusiness-isbn-perl
  libregexp-ipv6-perl libauthn-ntlm-perl debhelper snort-doc
```

`ip addr`

`sudo snort -T -c /etc/snort/snort.conf`

NAME: TANISHKA KUTWAL  
ROLL NO: T089

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```
mvl@LAPTOP-9N72DPF5:~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Setting up snort (2.9.20-0+deb11ubuntu1) ...
Snort configuration: interface default not set, using 'eth0'
Setting up liblw-protocol:~https-perl (6.13-1) ...
Setting up libwww-perl (6.76-1ubuntu0.1) ...
Setting up oinkmaster (2.0-4.2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
mvl@LAPTOP-9N72DPF5:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:7d:f0:e8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.254.23/20 brd 192.168.255.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe7f:f0e8/64 scope link
        valid_lft forever preferred_lft forever
mvl@LAPTOP-9N72DPF5:~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Running in Test mode

==== Initializing Snort ====
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [ 80:81 311 383 591 593 901 1220 1414 1741 1830 2381 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 4108 50002 55555 ]
PortVar 'SHELLCODE_PORTS' defined : [ 0:79 81:65535 ]
PortVar 'ORACLE_PORTS' defined : [ 1024:65535 ]
PortVar 'SSH_PORTS' defined : [ 22 ]
PortVar 'FTP_PORTS' defined : [ 21 2100 3535 ]
PortVar 'SIP_PORTS' defined : [ 5060:5061 5600 ]
PortVar 'FILE_DATA_PORTS' defined : [ 80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2381 2381 2809 3037 3128 3702 4343 4848 5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 4108 50002 55555 ]

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PortVar 'GTP_PORTS' defined : [ 2123 2152 3386 ]
Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsengine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
  Finished loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
```

sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf

```
mvl@LAPTOP-9N72DPF5:~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Preprocessor Object: SF_S7COMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: appid Version 1.1 <Build 5>

Total snort Fixed Memory Cost - MaxRss:105928
Snort successfully validated the configuration!
Snort exiting
mvl@LAPTOP-9N72DPF5:~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
Running in IDS mode

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  Search-Method = AC-Full-Q
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  Maximum pattern length = 20
Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsengine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
  Finished loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
```

Open the local rules file:

sudo nano /etc/snort/rules/local.rules

Add:alert icmp any any -> \$HOME\_NET any (msg:'ICMP Ping Detected';  
sid:1000001; rev:1;)

NAME:TANISHKA KUTWAL

ROLL NO: T089

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```
GNU nano 7.2 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any
(msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

```
ERROR: Can't start DAQ (-1) - No such device exists!
Fatal Error, Quitting..
mvl@LAPTOP-9N72DPFS:~$ sudo nano /etc/snort/rules/local.rules
mvl@LAPTOP-9N72DPFS:~$ ^C
mvl@LAPTOP-9N72DPFS:~$ sudo nano /etc/snort/rules/local.rules
mvl@LAPTOP-9N72DPFS:~$
```

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```
GNU nano 7.2 /etc/snort/snort.conf
#-----
# VRT Rule Packages Snort.conf
#
# For more information visit us at:
#   http://www.snort.org           Snort Website
#   http://vrt-blog.snort.org/     Sourcefire VRT Blog
#
# Mailing list Contact:   snort-users@lists.snort.org
# False Positive reports: fp@sourcefire.com
# Snort bugs:             bugs@snort.org
#
# Compatible with Snort Versions:
# VERSIONS : 2.9.20
#
# Snort build options:
# OPTIONS : --enable-gre --enable-mpls --enable-targetbased --enable-ppm --enable-perfprofiling --enable-zlib --enable-active-response --enable-normalize
#
# Additional information:
# This configuration file enables active response, to run snort in
# test mode -T you are required to supply an interface -i <interface>
# or test mode will fail to fully validate the configuration and
# exit with a FATAL error
#-----
#####
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder
# 3) Configure the base detection engine
# 4) Configure dynamic loaded libraries
# 5) Configure preprocessors
# 6) Configure output plugins
# 7) Customize your rule set
# 8) Customize preprocessor and decoder rule set
# 9) Customize shared object rule set
#-----
# Help      # Write Out  # Where Is    # Cut         # Execute     # Location    # M-U         # M-A         # M-]         #
# Exit      # Read File  # Replace     # Paste       # Justify     # Go To Line  # Undo        # Set Mark    # To Bracket  #
#           #           #            #            #            #            # Redo        # Copy        # Where Was   #
#           #           #            #            #            #            #            #            #            #            #
# 29°C     # Search     #            #            #            #            #            #            #            #
# Mostly cloudy
```

```
mvlu@LAPTOP-9N72DPFS:~$ sudo nano /etc/snort/snort.conf
mvlu@LAPTOP-9N72DPFS:~$
```

```
mvlu@LAPTOP-9N72DPFS:~$ sudo nano /etc/snort/snort.conf
mvlu@LAPTOP-9N72DPFS:~$ sudo snort -T -c /etc/snort/snort.conf
Running in Test mode

=== Initializing Snort ===
Initializing Output Plugins!
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Parsing Rules file "/etc/snort/snort.conf"
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Detection:
  Search-Method = AC-Full-Q
  Split Any/Any group = enabled
  Search-Method-Optimizations = enabled
  Maximum pattern length = 20
  Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsf_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
  Finished Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_appid_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dns_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_imap_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dnp3_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_gtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_s7complus_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_dce2_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_smtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsf_ssh_preproc.so... done
```

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sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf

```
mvl@LAPTOP-9N72DPF5: ~$ sudo snort -i enp0s3 -A console -c /etc/snort/snort.conf
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Tagged Packet Limit: 256
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  Finished loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsfe_appid_preproc.so... done
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsfe_dns_preproc.so... done
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsfe_imap_preproc.so... done
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsfe_dnp3_preproc.so... done
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsfe_gtp_preproc.so... done
  Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor/libsfe_s7complus_preproc.so... done
```

```
mvl@LAPTOP-9N72DPF5:~$ ping -c 4 192.168.254.23
PING 192.168.254.23 (192.168.254.23) 56(84) bytes of data:
64 bytes from 192.168.254.23: icmp_seq=1 ttl=64 time=2.59 ms
64 bytes from 192.168.254.23: icmp_seq=2 ttl=64 time=0.061 ms
64 bytes from 192.168.254.23: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 192.168.254.23: icmp_seq=4 ttl=64 time=0.044 ms

--- 192.168.254.23 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3058ms
rtt min/avg/max/mdev = 0.044/0.688/2.592/1.099 ms
mvl@LAPTOP-9N72DPF5:~$ sudo nmap -sS 192.168.254.23
```

sudo nmap -sS 192.168.254.23

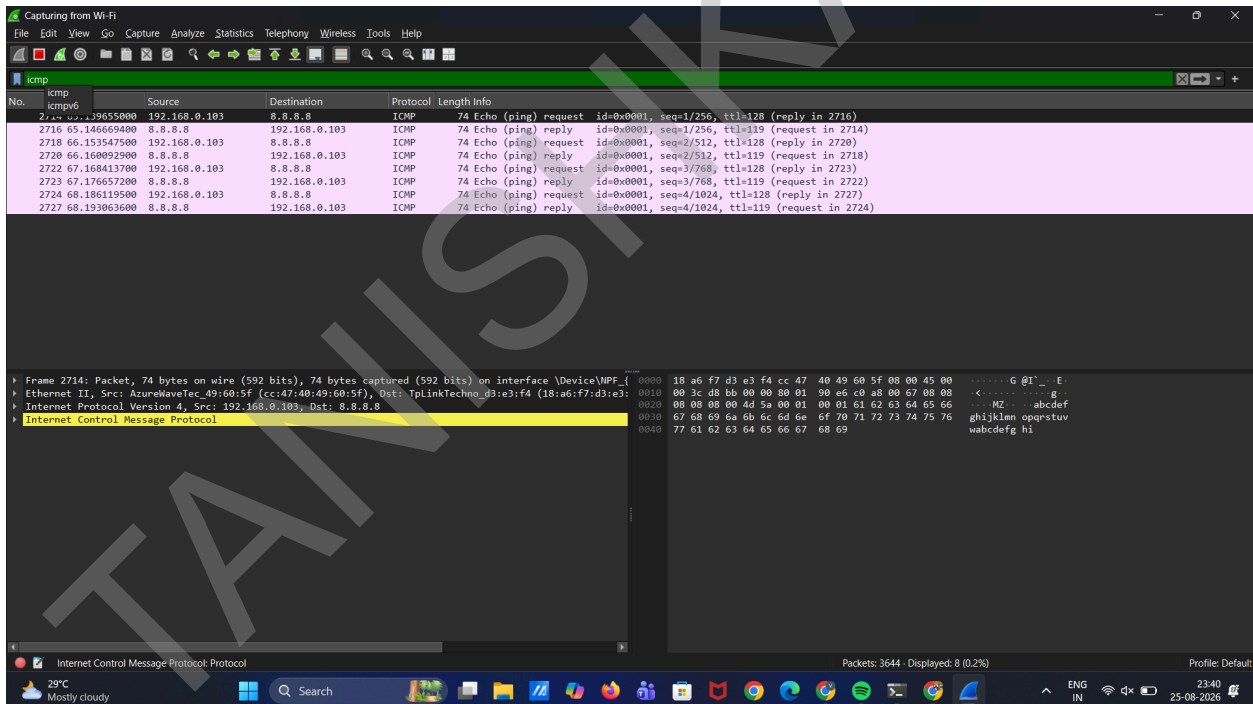
```
mvl@LAPTOP-9N72DPF5:~$ sudo nmap -sS 192.168.254.23
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-25 17:40 UTC
Nmap scan report for 192.168.254.23
Host is up (0.0000030s latency).
All 1000 scanned ports on 192.168.254.23 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
mvl@LAPTOP-9N72DPF5:~$
```

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```
mvl@LAPTOP-9N72DPF5: ~  
mvl@LAPTOP-9N72DPF5:~$ ping -c 4 192.168.254.23  
PING 192.168.254.23 (192.168.254.23) 56(84) bytes of data.  
64 bytes from 192.168.254.23: icmp_seq=1 ttl=64 time=0.302 ms  
64 bytes from 192.168.254.23: icmp_seq=2 ttl=64 time=0.056 ms  
64 bytes from 192.168.254.23: icmp_seq=3 ttl=64 time=0.052 ms  
64 bytes from 192.168.254.23: icmp_seq=4 ttl=64 time=0.068 ms  
  
--- 192.168.254.23 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3066ms  
rtt min/avg/max/mdev = 0.052/0.119/0.302/0.105 ms  
mvl@LAPTOP-9N72DPF5:~$
```



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